

Product Requirements Document: EcoPulse

Version: 1.0

Date: March 19, 2025

Author: Product Manager

1. Introduction

1.1 Vision

To empower organizations and their employees to take meaningful climate action by providing a seamless and engaging platform to track, understand, and reduce their carbon footprint, turning sustainability into a measurable business advantage.

1.2 Problem Statement

Employee-related activities, particularly commuting and business travel, are a significant and often untracked component of a company's carbon footprint (Scope 3 emissions). With increasing regulatory pressure (e.g., SEC climate disclosure rules) and financial incentives (e.g., Inflation Reduction Act), companies need an effective tool to measure these emissions, engage employees in reduction efforts, and unlock financial benefits. However, 84% of companies lack the tools to do so, leaving them exposed to financial penalties and missed opportunities.

1.3 Target Audience & Personas

Primary Persona: The "Eco-Conscious Employee" (e.g., Sarah, 32, Marketing Manager)

- **Goals:** Wants to live more sustainably but finds it difficult to track her impact and know what actions are most effective. She's motivated by personal values, and also by friendly competition and seeing her collective impact with colleagues.
- **Frustrations:** Lack of easy-to-use tools to track her carbon footprint across different activities (commute, work-from-home energy use, travel). Doesn't know how her small actions contribute to the company's larger goals.

Secondary Persona: The "Sustainability Officer" (e.g., David, 45, Head of ESG)

- **Goals:** Needs to accurately report on the company's Scope 3 emissions to comply with regulations and meet ESG targets. Wants to drive employee engagement in sustainability initiatives to create a greener company culture and enhance the brand's reputation.
- **Frustrations:** Manual data collection for emissions reporting is time-consuming and inaccurate. It's challenging to motivate all employees to participate in sustainability programs and to demonstrate the ROI of these initiatives to

leadership.

2. Goals and Objectives

Business Goals	User Goals (Employee)	User Goals (Admin)	Success Metrics (KPIs)
Achieve 50 paying corporate clients in the first year.	Easily track my daily carbon footprint.	Automate Scope 3 emissions reporting.	Monthly Active Users (MAU)
Secure a 15% reduction in carbon footprint for partner companies.	Get personalized tips to reduce my impact.	Increase employee engagement in sustainability.	Average % reduction in carbon footprint per user.
Establish EcoPulse as a leader in employee-driven sustainability.	Compete with colleagues on a leaderboard.	Identify potential cost savings and tax credits.	Number of corporate dashboards activated.

3. Features & Functionality

3.1 Feature Prioritization (RICE Framework)

The following features have been prioritized for the MVP based on the RICE (Reach, Impact, Confidence, Effort) framework to maximize value and minimize time-to-market.

Feature	Reach	Impact	Confidence	Effort	RICE Score
Employee Dashboard	5	5	5	3	8.3
Corporate Admin Dashboard	4	5	5	4	5.0
Automated Commute Tracking	5	4	4	3	5.3
Personalized Recommendations	5	4	5	3	6.7

ations					
Gamification (Leaderboards)	5	3	5	2	7.5

3.2 MVP Feature Set

Employee-Facing Application:

- **User Authentication:** Secure sign-up and login for employees.
- **Carbon Footprint Tracking:**
 - **Commute:** Manual entry and/or automated tracking of commute methods (driving, public transit, cycling, walking).
 - **Office & WFH:** Estimates based on office location and work-from-home days.
 - **Travel:** Manual entry for business travel.
- **Personalized Goals:** Ability to set and track monthly carbon reduction goals.
- **Action Recommendations:** AI-powered suggestions for reducing footprint (e.g., "Try carpooling twice a week to save X kg of CO2").
- **Progress Visualization:** Simple charts showing carbon footprint over time.
- **Activity Feed & Leaderboard:** View personal achievements and see how you rank against colleagues.

Corporate-Facing Dashboard:

- **Aggregated Analytics:** View the company's overall carbon footprint from employee activities.
- **Compliance-Ready Reports:** Generate reports for Scope 3 emissions.
- **User Management:** Onboard and manage employee accounts.

4. Design & User Experience

- **Clean & Intuitive UI:** The design should be simple, modern, and easy to navigate for non-technical users.
- **Data Visualization:** Charts and graphs should be clear and compelling, making it easy for users to understand their impact.
- **Gamification Elements:** Use badges, points, and leaderboards to drive engagement and friendly competition.
- **Accessibility:** Ensure the application is accessible to all users, including those with disabilities.
- **Dark Theme Support:** Provide a comfortable viewing experience in low-light

conditions.

5. Technical Stack

- **Backend:** Flask (Python)
- **Database:** PostgreSQL
- **Frontend:** HTML5, CSS3, JavaScript
- **Charts:** Chart.js
- **Authentication:** Flask-Login

6. Go-to-Market Strategy

- **Target Market:** Mid-to-large size companies in metropolitan areas with a strong commitment to ESG and sustainability.
- **Pricing:** Tiered subscription model based on the number of employees (\$9.99 - \$29.99 per employee/month).
- **Channels:**
 - Direct sales team focused on sustainability and HR leaders.
 - Partnerships with ESG consulting firms and industry associations.
 - Content marketing (blog posts, whitepapers) on the importance of Scope 3 emissions and employee engagement.

7. Future Roadmap (Post-MVP)

- **Automated Data Integration:** API integrations with utility companies, travel booking platforms, and ridesharing apps for fully automated tracking.
- **Advanced AI Recommendations:** More sophisticated AI to provide highly personalized and predictive recommendations.
- **Carbon Offsetting Marketplace:** Allow users and companies to purchase verified carbon offsets directly through the platform.
- **Mobile Application:** Native iOS and Android apps for an enhanced mobile experience.
- **Industry-Specific Modules:** Tailored features and benchmarks for different industries (e.g., tech, finance, healthcare).